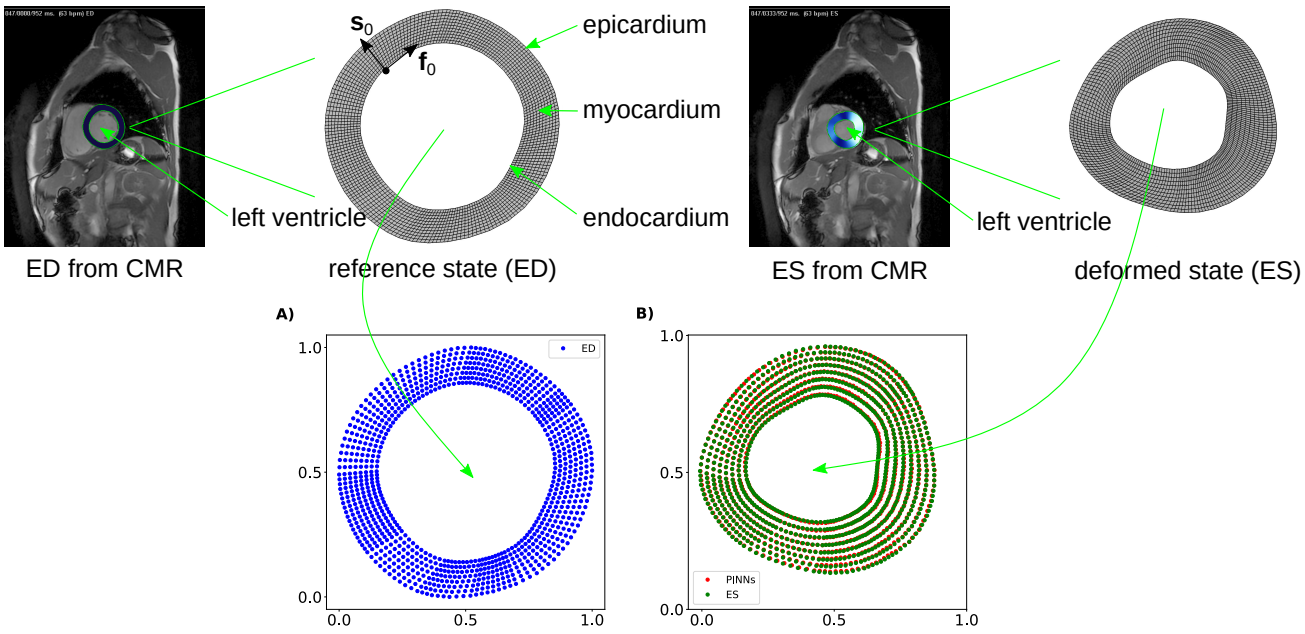
ORTA DOĞU TEKNİK ÜNİVERSİTESİ
MIDDLE EAST TECHNICAL UNIVERSITY

Computational Solid Mechanics Lab (CSML)
Department of Civil Engineering
Middle East Technical University (METU)

Master Thesis (4)

Physics-informed Surrogate Modeling of Isotropic and Anisotropic Hyperelasticity

Description



cMRI data showing the left ventricle short axis view along with the corresponding finite element geometry and mesh reconstruction for the end-diastolic (ED) with the representative fiber $\mathbf{f}_0$ and sheet $\mathbf{s}_0$ orientations and the end-systolic stage (ES). PINNs prediction of deformation configuration of the annulus from A) ED stage (undeformed configuration) to B) ES stage (deformed configuration).

There is an ever growing interest in physics-informed neural networks (PINNs) and graph-neural networks (GNNs) in predicting the mechanics of solids and fluids, e.g., from hyperelasticity, thermo-mechanics, damage to plasticity. These surrogate modeling approaches emphasize their utility in both forward and inverse simulations (material characterization). Physics-aware models essentially adhere to the governing equations and the initial/boundary conditions of the system by integrating physical laws into the neural network architecture. Traditional methods for simulating the mechanics of materials, though providing high-fidelity outcome, often rely on complex and computationally expensive finite element models. In contrast, physics-informed surrogate models enable the accurate modeling of mechanical response without relying heavily on large datasets, overcoming the limitations of ANNs and decreasing the overall computational costs associated with FEM once training is completed.

In this project, we aim to **develop and implement surrogate models that incorporate physical laws directly into the neural network framework**, providing a more efficient and potentially more accurate simulation of hyperelasticity in 2D and 3D. The expected outcomes include

- a predictive computational tool validated against physical data capable of forward and inverse analysis
- to lay the foundation for future studies in the fields of biomedical and civil engineering

Tasks

- literature study regarding surrogate modeling of arterial wall mechanics in PINNs and GNNs
- building up machine learning framework and adding physical laws considering local and global aspects of computational mechanics
- performing training & validation runs on benchmark problems for isotropy and anisotropy
- preparation of research findings for journal publications

Your qualifications

- B.Sc. degree holder in Civil Engineering, Mechanical Engineering, or related discipline
- strong motivation to learn computational solid mechanics and machine learning
- programming experience in Python and familiarity with PyTorch and/or TensorFlow are two pluses
- good command of written and spoken English

We offer

- direct research mentorship and guidance throughout the thesis work
- access to computational infrastructure and research facilities

Contact

Interested candidates should send their CV & Letter of Interest to:

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